Image Processing and Medical Imaging

Image Registration

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Image Registration

- matching two images so that corresponding coordinate points in the two images correspond to the same physical region of the scene being imaged
- also referred to as **image fusion**, **superimposition**, matching or **merge**

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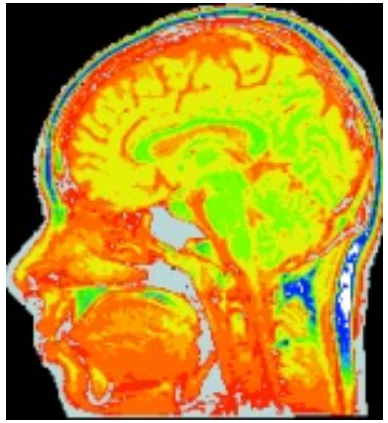
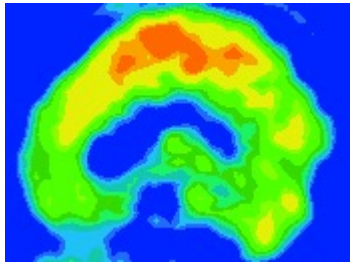
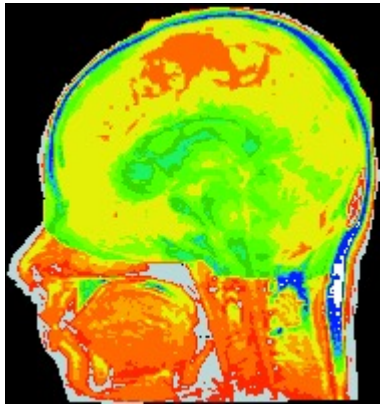
Example



MR



SPECT



Registered

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Source: www.medischebeeldvorming.nl

Applications

- Diagnosis
  - Combining information from multiple imaging modalities
- Studying disease progression
  - Monitoring changes in size, shape, position or image intensity over time
- Image guided surgery or radiotherapy
  - Relating pre-operative images and surgical plans to the physical reality of the patient
- Patient comparison or atlas construction
  - Relating one individual's anatomy to a standardized atlas

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Classification of the methods

- Dimensionality
  - 2D-2D, 3D-3D, 2D-3D
- Nature of registration basis
  - Image based
    - Extrinsic, Intrinsic
  - Non-image based
- Nature of the transformation
  - Rigid, Affine, Projective, Curved
- Interaction
  - Interactive, Semi-automatic, Automatic
- Modalities involved
  - Monomodal, Multimodal, Modality to model

- Subject:

  Intrasubject

  Intersubject

  Atlas

- Domain of transformation
  - Local, global
- Optimization procedure
- Object

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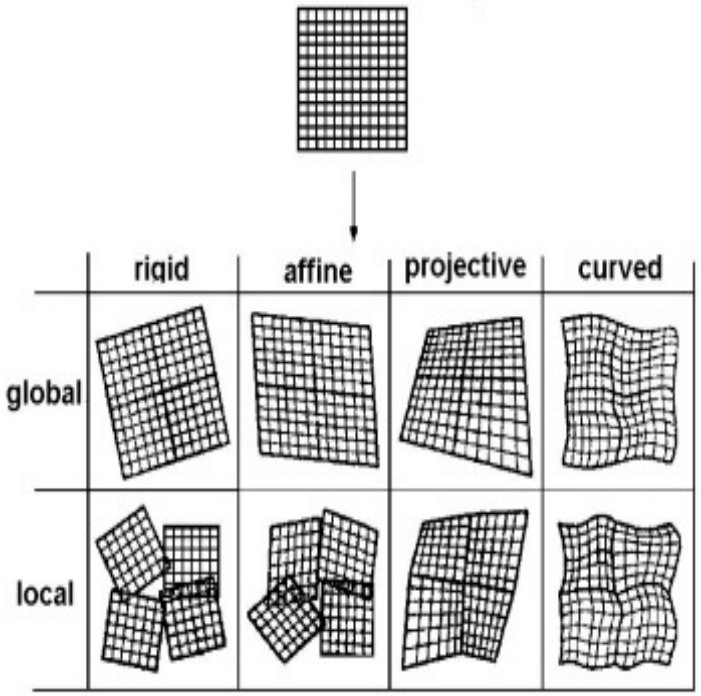
Transformation

Relates the position of features in two images

- **Rigid**
  - translations and rotations
- **Affine**
  - Also allows scaling and shearing
- **Curved**
  - Allows the mapping of straight lines to curves
- **Perspective**
  - The parallelism of lines need not be preserved



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Source: https://www.researchgate.net/figure/1-les-differents-modeles-de-transformations-geometriques_fig18_37813631

Behind the Registration Algorithms

Rigid & affine

- Landmark based
- Edge based
- Voxel intensity based
- Information theory based

Non-rigid

- Registration using basis functions
- Registration using splines
- Physics based (Elastic, Fluid, Optical flow)

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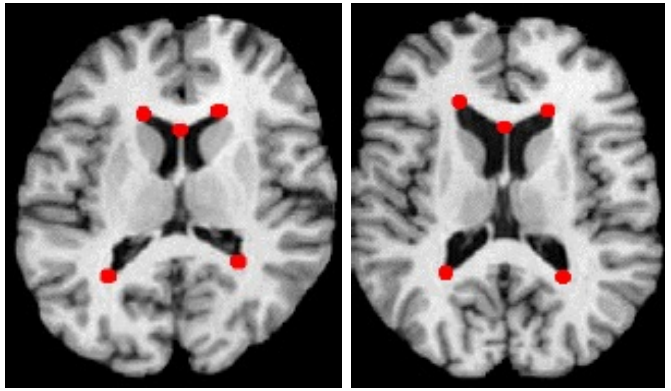
Landmark Based Method

- Identifying corresponding points in the images and inferring the image transformation.
- Types of landmarks
  - Extrinsic
    - artificial objects attached to the patient
  - Intrinsic
    - internal anatomical structures



- Computing the average or "centroid" of each set of points → translation.
- Rotated this point set about the new centroid until the sum of the squared distances between each corresponding point pair is minimized.

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Source: ttps://www.researchgate.net/figure/Mean-brain-volumes-for-young-and-older-subject-groups_tbl1_5336810

Surfaced Based Method

- Method
  - Extracting corresponding surfaces
  - Computing the transformation by minimizing some measure of distance between the two surfaces
- Algorithms used
  - The “Head and Hat” Algorithm
  - The Iterative Closest Point Algorithm

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Voxel Intensity Based

- Method
  - Calculating the registration transformation by optimizing some measure calculated directly from the voxel values in the images.
- Algorithms used
  - Registration by minimizing intensity difference
  - Correlation techniques
  - Ratio image uniformity
  - Partitioned intensity uniformity

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Information Theory Based

- Image registration is considered as to maximize the amount of shared information in two images, reducing the amount of information in the combined image.
- Algorithms used
  - Joint entropy
    - Joint entropy measures the amount of information in the two images combined.
  - Mutual information
    - A measure of how well one image explains the other, and is maximized at the optimal alignment.
  - Normalized Mutual Information

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Registration Using Basis Functions

- Represent the deformation field using a set of basis functions
  - Fourier (trigonometric) basis functions or wavelet basis functions.
  - Implement smoothness constraint by linear combination of basis functions.
  - The trigonometric basis functions corresponds to a spectral representation of the deformation field where each basis function describes a particular frequency of the deformation.

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Registration Using Splines

- Let us suppose set of corresponding points or landmarks (control points) can be identified
- At control points, interpolate or approximate the displacements to map the location of the control points in both images
- Between control points, they provide a smoothly varying displacement field

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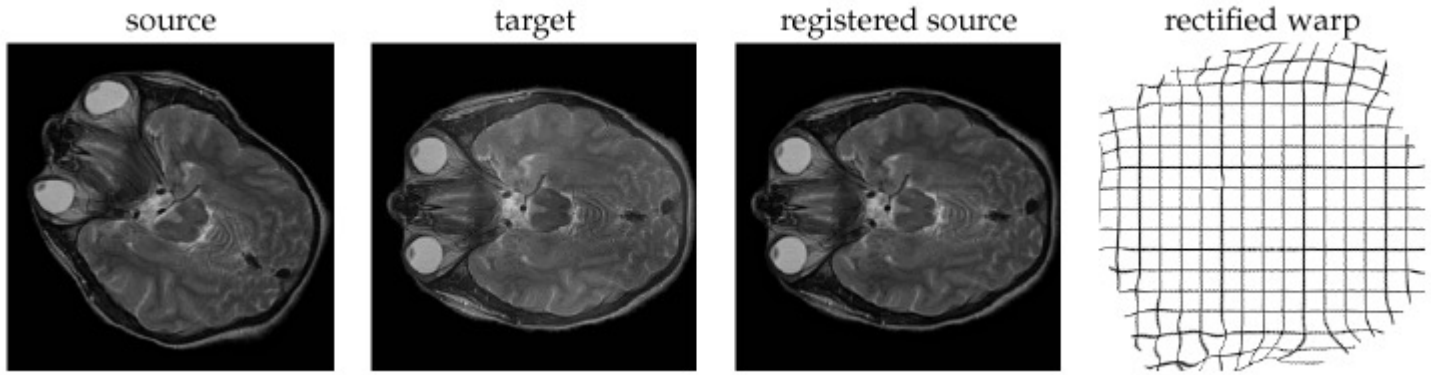
Elastic Registration

- Model the deformation as a physical process resembling the stretching of an elastic material
  - The physical process is governed by the internal force & external force.
  - Described by the Navier linear elastic partial differential equation.
- The external force drives the registration process
  - The external force can be the gradient of a similarity measure (local correlation measure based on intensities, intensity differences or intensity features such as edge and curvature).
  - Or the distance between the curves and surfaces of corresponding anatomical structures.



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Other Physics Based Registration

- Fluid registration
  - The image was modeled as a highly viscous fluid.
- Registration using mechanical models
  - Using model to simulate the properties of rigid, elastic and fluid structures.
- Registration using optical flow.

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Optimization

- Many registration algorithms require an iterative approach
  - The initial estimate of the transformation is gradually refined.
  - In each iteration, the current estimate of the transformation is used to calculate a similarity measure.
  - Makes another estimate of the transformation, evaluates the similarity measure again, and continues until the algorithm converges.

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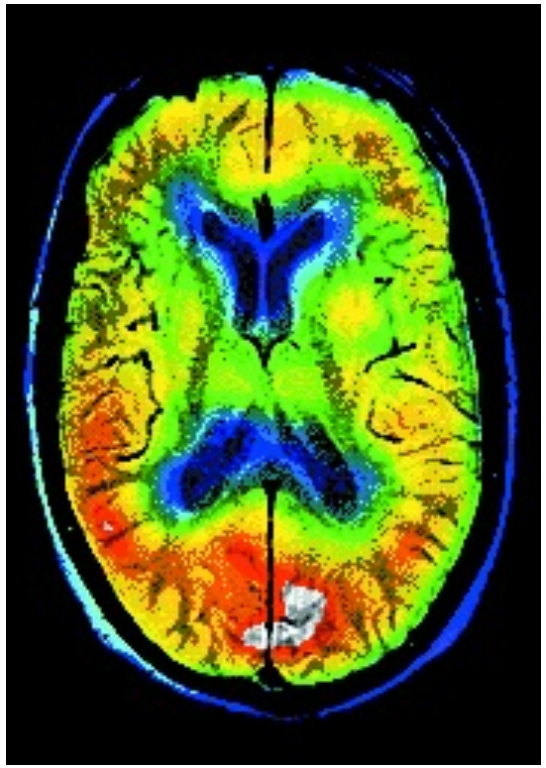
Visualization

- Color overlay
- Interleaved pixel or chessboard fusion
- Dynamic alternating display
- Split view displays
- Subtraction images
- Etc.



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Thank you for your attention